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2026-03-25

neev
ACADEMY

Summative Assessment - Science

Unit- 4

Forces at Work: How Simple Machines Transform Energy?

MYP Year - 3

Grade: 7

Name of the student- Ahmed Shaker

Grade and section - 7A

Statement of Inquiry: Simple machines transform energy and change how forces function in performing work, shaping the innovations that impact our daily lives. [

Key Concept: Change

Related Concepts: Function, Transformation

Global context: Scientific and technical innovation

MYP Assessment Objective: Criteria D

Time Allotted: 55 minutes

Assessment Type:

Formative	
Summative	*

Moderated
D-4

B

Sailor

Source A: The Problem – Physical Strain in Healthcare

In modern hospitals, nurses and healthcare assistants often lift, turn, or move patients who have limited mobility. These tasks involve strong pushing and pulling forces.

The human body works like a system of levers:

- Pivots (Fulcrums): Joints such as the elbow, shoulder, and lower back
- Levers: Bones
- Effort: Muscles that create movement

When a nurse bends to lift a patient, the lower back acts as a pivot. Because the load (the patient) is far from the pivot, the muscles must apply a large amount of effort. Over time, this repeated force can cause fatigue, back injuries, and joint damage.



Source B: The Science of Exoskeletons

An exoskeleton is a wearable machine that supports the body.

- Passive exoskeletons use springs to store energy when bending and release it when lifting.
- Active exoskeletons use motors and sensors to assist movement.



These devices act as an external support frame. They absorb some of the force that would normally act on the body, allowing tasks to be completed with less effort and improving efficiency.



Source C: Impact Data & Feedback

Researchers studied the use of exoskeletons in a hospital for six months.

Factor	Observation / Data Collected
Physical Impact	40% reduction in muscle activity; 25% fewer strain injuries
Economic	Costs increased by 15% due to maintenance and batteries
Social / Interaction	60% of patients felt nurses seemed less approachable
Operational	Setup took 10 minutes at the start of each shift
Environmental	Batteries need daily charging and replacement every 2 years

Use information from all sources to develop your own explanation and evaluation.

Task

Discuss and evaluate how exoskeletons apply scientific principles to solve problems in healthcare, and analyse their wider implications.

Write a structured response (500–550 words).

Your response should address the following:

4-

exoskeletons — Science — Unit 3 (4 if you count from 1) — Section D

The human body is simply a complex machine. (Complex in the sense that they contain multiple simple machines, ~~not~~ not complex in the sense that it is a real machine a and an imaginary machine b in the form $a+bi$.) This proposition might seem heretic to some — probably people who don't like it when humans are reduced to machines. However, these machines present in our body may not be suited to the task.

Very long introduction

As ~~an~~ a scientifically-unrelated but capability-related example, we cannot visualize time as fluid, or length as fluid, or different. This ~~is~~ — a fundamental concept in special relativity — only becomes relevant at velocities close to c . ~~Classical~~ physics was evolved to assume that time is a constant, and velocity addition is linear. This happened because we travel at ~~$v \ll c$~~ , which is why it never occurred to us.

not required

Diminuf, we weren't meant to do certain tasks. We weren't meant to lift beds with people on them and equipment. This is where exoskeletons come in.

Exoskeletons, by definition, are a skeleton that is present on the outside of an animal's body — some insects have retrol exoskeletons, like ^{certain} beetles — in contrast with an internal endoskeleton. In this context, the term refers to artificial skeletons worn by humans in order to do their work. Especially in healthcare.

People in hospitals often need help. For patients and nurses to provide this help, they need to transport a person. It is not an easy task to move a person. The average adult human has a mass of around 70 ± 10 kg. On earth, that amounts to a force of around 700 ± 100 N. To counteract this, ^{one} ~~you~~ must apply the same force away from earth in order to keep the load stable. Humans can do this, but it is not optimal.

There are 2 main types of exoskeleton: passive and active. Passive exoskeletons

* Please read the q to write based on that.

rely on stored energy:

Relevant but lacks in-depth explanation

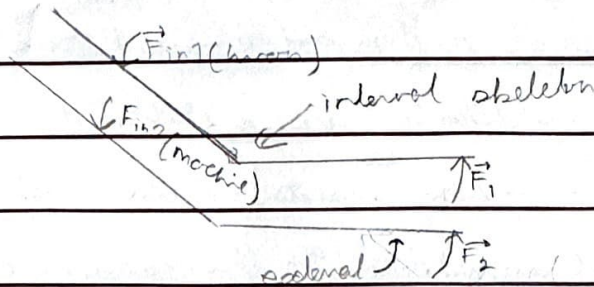
← stretched spring

← compressed spring

The springs are compressed when whilst being assisted by the earth's gravity, and released when working against it. This works surprisingly well, and it is not ~~are~~ very expensive. The downside, however, is that it only applies to certain tasks, such as bending and getting up. Also, due to conservation of energy, you can't get more out of than your maximum potential.

The next type is on active exoskeleton. This relies on ~~power~~ motors and sensors. The sensors detect movement, which the motors then amplify. This

is a diagram:



in this case, the total force is $\vec{F}_1 + \vec{F}_2$ with just a known input of $\vec{F}_{in1}$. OK, yes, the length of the ~~lever~~ which length are you referring to? does determine the force needed — specifically, it is:

$$d_1 \vec{F}_1 = d_2 \vec{F}_2 \quad (\text{not related to the previous diagram})$$

but if we assume the distance is ~~constant~~ ^{and the same}, then the new force output is greater than the single input. Why? This is why:

Discussion about body levers needed.

Advantages: scientific explanation but unclear

Name : _____

Date : _____

- implications only stated not discussed
- Data based evidence limited - (analysis)

$$d_1(\vec{F}_1 + \vec{F}_2) = d_2(\vec{F}_{in1} + \vec{F}_{in2})$$

The distances are constant and the same, so: → explain what each term stands for.

$$d(\vec{F}_1 + \vec{F}_2) = d(\vec{F}_{in1} + \vec{F}_{in2})$$

$$\vec{F}_1 + \vec{F}_2 = \vec{F}_{in1} + \vec{F}_{in2}$$

$$\vec{F}_1 + \vec{F}_2 - \vec{F}_{in2} = \vec{F}_{in1}$$

$$\therefore \vec{F}_{in1} \leq \vec{F}_1 + \vec{F}_2$$

■ QED

There are numerous advantages to this. Because we are now no longer limited by ~~the~~ ~~patient's~~ human strength, we can abuse this power. If we take $\vec{F}_{in1} = 0$, then we can let the machine do all of the work. Therefore, this is not an assistant — it is, rather, an object that performs tasks on its own.

What is the impact? There are actually, none then one. For instance, there has been a 25% decrease in injuries from strained muscles (C). This is ongoing. However, there are some issues. Exoskeletons are designed to be efficient and not aesthetically pleasing. Due to this, 60% of patients thought nurses looked less approachable (C), ruining the friendly atmosphere — actually, why does that even matter? People go to hospitals to help other people or get help, not make new friends. Also, who are these 60% that care about social life? I'm with those 40%, wherever they are.

Aside from that, this solution is not very economically feasible. Costs went up by 75% (C). Public hospitals don't have a problem, as they can just increase taxes — but the problem is with private ones.

~~also~~ You have used scientific content which not completely relevant. Implications have not been discussed properly at all. Please read the question properly.